LAB 3 – ARCHITECTURE SELECTION & JUSTIFICATION

System: Smart Parking Space Rental & License Validator
Selected Architecture: Layered Architecture

Architecture Selection

We chose **Layered Architecture** for the Smart Parking Space Rental & License Validator system.

Reason 1 – Separation of Concerns

Layered Architecture provides a clear separation between the user interface, parking-management logic, vehicle validation, billing operations, and data storage. This makes the system easier to understand, maintain, and modify.

Reason 2 – Suitability for the System

The Smart Parking system has clearly defined functions such as parking-bay reservation, vehicle number-plate validation, parking fee calculation, overstay detection, and parking-bay availability management. Organizing these functions into logical layers provides a simple and structured design.

Security Advantage

The layered design keeps security-sensitive vehicle validation and transaction-processing operations behind the business layer instead of exposing them directly through the user interface. This supports controlled access to reservation and validation data. The system also requires invalid vehicle validation attempts to be handled securely. Exception_Flow (1)

Performance Benefit

The architecture separates user interaction from business processing and data storage, allowing the vehicle-validation functionality to be optimized independently. This supports the requirement that parking-reservation validation be completed in **under 300 ms** while satisfying the required security checks.